

InfinityProductOS

Functional User Guide

*Enterprise Payment Workflow Platform • Built Jun 12–22,
2026*

*AI Developers: Gemini (Foundation) + Claude (Integration &
Runtime)*

Stat	Value
Total commits	107
Studios built	15+
ISO 20022 fields	3,013
Workflow templates	35+
Transaction lifecycle states	12
Workflow steps wired	137 (100%)

1. Platform Overview

InfinityProductOS is an enterprise payment workflow platform built on a single principle: no business logic is hardcoded. Every rule, workflow graph, formula, API configuration, and screen definition is stored as structured data and interpreted at runtime by stateless execution engines. Changes take effect the moment you save them — no code deployment needed.

1.1 The 8-Layer Architecture

Layer	Name & Purpose
L0 — Physical Edge	Devices (POS, mobile, IoT). Data producers and consumers only.
L1 — Visual Studios	React canvas studios where business users author logic: rules, workflows, calculations, screens.
L2 — Agentic AI	Infinity AI assistant — NLP prompt-to-canvas, image-to-workflow, behavioral AI.
L3 — Semantic Bloodstream	ISO 20022 Field Registry — 3,013 fields forming the universal vocabulary.
L4 — Deterministic Execution	FastAPI + Python engines. Runs logic authored in L1 against real transaction data.
L5 — Persistent Storage	SQLite (local dev) / PostgreSQL (production). Immutable evidence ledger.
L6 — Governance & Compliance	PII masking, 4-Eye approval checks, OIDC auth, append-only audit log.
L7 — Global Isolation	Multi-tenant, multi-region, multi-currency, multi-language isolation.
L8 — Fault Tolerance	Celery async checkpointing, atomic DB transactions, circuit breakers on all external APIs.

1.2 How the Engines Connect

- You author a workflow in the Workflow Designer. It saves as a JSON graph in the database.

- You author rules in the Business Rules Studio. They save as JSON conditions with token codes.
- You author formulas in the Calculation Engine. They save as symbolic expressions.
- At runtime, a payment hits the Workflow Executor. It walks the graph node by node.
- At each node, the executor invokes: Business Rule Engine + Calculation Engine + API Dispatcher + Event Bus.
- Results update the WorkflowExecutionInstance. The Transaction Workflow Screen polls and renders the metro tracker in real time.

1.3 Package System

Everything is scoped to a Product Package (e.g. "Payment Hub"). Each Package contains Products, and each Product can have Sub-Products. The Two-Key Cockpit Lockdown enforces that both Package AND Product must be selected before designer studios activate.

How to set context: Click your package name in the top-left corner. Select a Package, then a Product. The lock banner turns green when both are set.

2. Navigation Guide

2.1 Master Data

Menu Item	What it contains
ISO Field Registry	All 3,013 ISO 20022 standard fields — the vocabulary for every studio.
Packages	Create and manage Product Packages (top-level tenants).
Products	Products within a package, each with its own studio context.
Sub-Products	Third level of the hierarchy for granular product variants.
Roles & Users	User role definitions (Admin, Designer, Operator, Auditor, Risk).
Authorization Matrix	Which roles can access which studios and actions.
Access & Authorization	Entitlement configuration: rule-based access policies per product.

2.2 Designer Studio

Studio	What you build here
ISO Field Registry	Browse fields, set display preferences, manage PII flags.
Screen Designer	Build data-entry and review screens bound to ISO fields. Three types: Input, Review, Display.
Workflow Designer	Draw payment workflow graphs. Add nodes, connect with condition arrows, set SLAs.
Business Rules Studio	Author IF-THEN rules. Publish with token codes for workflows to reference.
Calculation Engine	Define symbolic math formulas (FX conversion, net settlement, fees).
API Designer	Configure external API connections with rate limits, circuit breakers, PII masking.

Report Designer	Build analytics dashboards with KPI cards, charts, and data grids.
Reconciliation Engine	Define match rules for nostro/vostro account reconciliation.
File Template Designer	Define fixed-width or delimited file formats for inbound payment files.
Data Gateway Mapper	Map incoming file fields to ISO 20022 standard fields.
Notification Engine	Author email, SMS, and letter templates.
Document Checklist Canvas	Build compliance document checklists per product.
Unstructured Document Studio	Onboard legacy scanned documents with OCR and AI extraction.
Integration Gateway	API Gateway Designer and Batch Gateway Designer for external system connectivity.
Legacy Screen Onboarding	Import and digitize legacy mainframe screens using OCR + Anthropic Vision.

2.3 Runtime Operations

Menu Item	Purpose
Transaction Workflow Screen	Watch live payments move through workflow nodes. Take actions: approve, reject, retry, reverse.
File Import Gateway	Upload and execute inbound payment files against live workflows.
Reconciliation Tracker	Monitor nostro/vostro match runs and reconciliation parity.
Event Catalog	Inspect system event logs and audit streams.
Execution Traces	Deep-dive trace view for individual transaction node executions.
Global Technical Dashboard	System telemetry: backend health, API connectivity, field registry, Celery worker.

3. ISO Field Registry

The ISO Field Registry is the semantic backbone of the platform. Every field in every studio — rule conditions, formula variables, screen components, mapper columns, report dimensions — is anchored to a field here.

3.1 What it contains

- 3,013 ISO 20022 standard fields across all message types (pacs, camt, pain, admi)
- Each field has: ISO name, client display name, data type, PII flag, domain taxonomy
- Fields organized by domain: Payment, Settlement, Compliance, Customer, Reference Data

3.2 Actions

Action	How
Search fields	Type in the search box — debounced server-side search across all 3,013 fields instantly
Filter by type	Use the type filter dropdown (String, Decimal, Date, Boolean, etc.)
Filter PII fields	Toggle the PII filter to see only fields containing personal data
Set display preference	Click a field row, toggle between ISO Standard Name and Bank Custom Name
Filter by domain	Use the domain taxonomy filter to see fields for a specific payment message type

4. Workflow Designer

Draw the step-by-step graph of how a payment moves through your organization — from ingest through validation, enrichment, approval, and settlement.

4.1 Canvas basics

- Click + on any node to add a connected next node
- Drag nodes to reposition on the canvas
- Click an arrow/edge to add a condition label (e.g. "Amount > \$10,000")
- Click a node to open the Properties Drawer on the right
- Toggle between Flow View and Swimlane View (grouped by team/participant)
- Save Draft to persist without activating, or Publish to make the workflow live

4.2 Node types (21 canonical types)

Node Type	What it does
START (green)	Entry point — every workflow begins here
END (red)	Terminal node — workflow completes here
BUSINESS_RULE (blue)	Evaluates an IF-THEN rule at this step. Pick rule from token code dropdown.
CALCULATION (purple)	Runs a symbolic formula. Result stored in execution context.
API_CALL (amber)	Calls an external API (SWIFT GPI, FedNow, OFAC screening, etc.)
HUMAN_APPROVAL (teal)	Pauses the workflow. A human operator must Approve or Reject to continue.
EMIT_EVENT (orange)	Broadcasts an event code to the event bus (e.g. EVT_PAYMENT_SETTLED)
SUB_WORKFLOW (indigo)	Executes another workflow as a nested child. Supports infinite nesting.
DECISION_GATEWAY (diamond)	Branches the flow based on a condition. Routes to different next nodes.

RECONCILIATION (grey)	Triggers a nostro/vostro reconciliation run at this step.
REPORT (light blue)	Generates and attaches a report to the transaction record.
NOTIFICATION	Sends an email/SMS/letter using a Notification Engine template.

4.3 Node Properties Drawer

Click any node to set: title, step type, orchestration steps (rule → calc → API → event sequence), SLA duration, failure handling (RETRY/HALT/SKIP/COMPENSATE), reversal recipe, and swimlane participant.

4.4 ISO 20022 Template Library (35+ templates)

Category	Templates included
SWIFT	MT103 Cross-Border, MT202 Bank Transfer, pacs.008, pacs.009
FedNow	Instant Credit Transfer, Request for Payment, Return
RTP	Real-Time Payment Initiation, Return, Recall
CHIPS	Funds Transfer, Bilateral Settlement
SEPA	Credit Transfer (SCT), Instant (SCT Inst), Direct Debit
ACH	Batch Credit, Batch Debit, IAT, Return, NOC
Cash Management	camt.052 Intraday, camt.053 Statement, camt.054 Notification

4.5 Image-to-Workflow (AI feature)

Click the camera icon in the toolbar. Paste or upload a screenshot of any process diagram (Visio, PowerPoint, whiteboard photo, Lucidchart export). The AI parses it and populates the canvas with nodes and edges automatically.

4.6 Wiring Audit

Click the link icon (Wiring Audit) in the toolbar to open the Wiring Audit panel. It scans every workflow node and shows any step with a type but no target — meaning it would fire as a silent no-op. Select targets from dropdowns and click Apply Wiring to fix all at once. A green checkmark means all steps are fully wired.

5. Business Rules Studio

Author IF-THEN rules that the Workflow Executor evaluates at BUSINESS_RULE nodes. Rules are stored as JSON and evaluated at runtime — changing a rule takes effect on the next transaction, no deployment needed.

5.1 Rule Actions

Action	Effect
APPROVE_STEP	Marks node as passed. Workflow continues to next node.
BLOCK_PAYMENT / REJECT_STEP	Halts entire workflow. Instance status becomes REJECTED.
FLAG_FOR_REVIEW	Adds a flag and pauses for operator review.
CANCEL_TRANSACTION	Voluntarily terminates workflow. Status becomes CANCELLED (distinct from REJECTED).
REQUIRE_4EYE	Triggers mandatory 4-Eye human approval at this point.
EMIT_EVENT	Broadcasts an event code to the event bus.

5.2 Live Rules (Golden Path)

Token	What it checks
BRE-XBDR-AML-HVT-V1	AML high-value threshold. Flags transactions over \$500,000 for mandatory 4-Eye review.
BRE-XBDR-OFAC-SCRN-V1	OFAC sanctions screening. Checks beneficiary against sanctions lists. Blocks if match found.
BRE-XBDR-FX-STALE-V1	FX rate staleness. Rejects if FX rate is more than 4 hours old.

6. Calculation Engine

Define symbolic mathematical formulas that the Workflow Executor resolves at CALCULATION nodes. All arithmetic uses exact Decimal precision — no floating-point rounding errors in financial calculations.

6.1 Live Formulas (Golden Path)

Token	Formula	What it calculates
FX_CONVERTED_AMOUNT	$\text{INSTRUCTED_AMT} \times \text{FX_RATE}$	Converts payment amount to settlement currency
NET_SETTLEMENT_AMOUNT	$\text{GROSS_AMOUNT} - \text{CORRESPONDENT_FEE}$	Net amount after deducting correspondent bank fees
CORRESPONDENT_FEE	$\text{GROSS_AMOUNT} \times \text{FEE_RATE}$	Correspondent bank fee as percentage of gross amount
DAYS_OUTSTANDING	$\text{TODAY} - \text{VALUE_DATE}$	Number of days a payment has been outstanding

7. API Designer

Configure external API connections that workflow nodes can invoke. Every configuration enforces rate limiting and circuit breaking.

7.1 Live API Configurations

API	Purpose
SWIFT GPI Tracker — Submit Payment	Submits payment to SWIFT gpi network, receives UETR tracking reference
Bank of England RTGS — Settlement Confirmation	Confirms final settlement via Bank of England RTGS system
OFAC SDN Screening API	Screens beneficiary against OFAC Specially Designated Nationals list
FedNow Real-Time Settlement	Submits instant payments to Federal Reserve FedNow service
Open Exchange Rates Feed	Fetches live FX rates for currency conversion
Refinitiv World-Check Screening	Enhanced sanctions and PEP screening

8. Transaction Workflow Screen

The operations center of the platform. Everything you designed in the studios comes alive here. Watch payments move through your workflow in real time, take action on paused transactions, search across millions of records, and manage failed reversals.

How to get there: Runtime Operations → Transaction Workflow Screen

8.1 The Metro Tracker

A London Underground-style station map. Each station = a workflow node. Payments progress left-to-right as they execute.

Color	Meaning
Green (solid)	Step completed successfully
Blue (pulsing)	Currently executing right now
Amber (pulsing)	Paused — waiting for human approval or manual action
Red	Blocked or failed — needs operator intervention
Grey	Not yet reached
Purple	Reversed — compensation completed

8.2 The 12 Transaction Lifecycle States

State	Meaning	What the operator sees
RUNNING	Actively executing	Blue pulsing station. No action needed.
PAUSED	Waiting for 4-Eye approval	Amber pulsing. Approve or Reject buttons active.
COMPLETED	All steps done	All stations green. Full execution trace visible.
REJECTED	A rule blocked it	Red terminal. Reason shown in trace (which rule, which condition).

BLOCKED	Technical error	Red station. Retry button active. Error detail in step-issue panel.
CANCELLED	Voluntarily stopped	Grey terminal. Shows who/what cancelled and reason code.
RETRYING	Failed step being retried	Amber station. Shows retry count and next attempt time.
AWAITING_REPAIR	Assigned to repair queue	Amber station. Shows assigned repair agent name.
FAILED_TECHNICAL	Unrecoverable failure	Red terminal. Requires manual escalation.
REVERSED	Saga compensation done	Purple stations. All compensating actions completed.
REVERSING	Compensation in progress	Purple pulsing.
REVERSAL_FAILED	Compensation failed	Red terminal. Lands in Reversal Recovery Queue.

8.3 Action Buttons

Button	When available	What it does
Approve	Status is PAUSED	Resumes the workflow from the paused node immediately.
Reject	Status is PAUSED	Terminates workflow as REJECTED. No further nodes run.
Retry	Status is BLOCKED	Re-executes failed step. Increments retry counter (max 3).
Cancel	Any active status	Voluntarily stops workflow. Status becomes CANCELLED.
Reverse	Any completed step	Opens Reversal Drawer to initiate saga compensation.

Keyboard shortcuts: A = Approve, R = Reject, Escape = close panels.

8.4 Running a New Transaction

Click the green **Play (Run)** button in the header. Fill in: Workflow, Amount, Currency, Beneficiary BIC, Beneficiary Name, FX Rate, Value Date. Click Execute Transaction. The golden path (\$592,500 USD → BARCGB22) pauses at NODE-02 (AML check) for 4-Eye approval, then completes via SWIFT GPI and Bank of England RTGS.

8.5 Transaction Search

Search across all instances by: amount, currency, beneficiary name, BIC, status, workflow name, date range, or instance ID.

8.6 Bulk Operations

Select multiple transactions via checkboxes, then: Approve all (PAUSED), Retry all (BLOCKED), or Cancel all (with bulk reason code).

8.7 Reversal Recovery Queue

When a reversal fails, the transaction lands in REVERSAL_FAILED state and appears here. View error details, filter by assigned/unassigned, click View to load the full instance in the metro tracker for manual intervention.

9. Report Designer

Widget type	Use case
KPI Card	Single metric with trend indicator (e.g. Total Settlements Today: 1,247)
Bar Chart	Compare values across categories (e.g. volume by currency)
Line Chart	Track metrics over time (e.g. 30-day settlement trend)
Data Grid	Tabular view with column sorting and filtering
Pie / Donut	Proportional breakdown (e.g. transactions by status)
Gauge	Single metric against a target (e.g. SLA compliance rate)

10. Reconciliation Engine

Define match rules that compare internal records (nostro) against correspondent bank records (vostro) to confirm settlement parity. The golden path template "Nostro vs Vostro Daily Matching" uses 4 rules: exact UETR match, amount match within 0.01 tolerance, value date within T+1, and exact currency code match.

11. Upcoming Features (E7)

Feature	What it will do	Status
Entitlements Enforcement	Operators see only their team's transactions. Enforced at the API level via X-User-Role header.	In progress (Gemini)
Event Fan-Out	When a workflow emits EVT_PAYMENT_SETTLED, downstream systems (Insights, Reconciliation, Behavioral AI) automatically react.	Planned
Mobile Metro Tracker	The station map scales for narrow screens and mobile devices.	Planned

Appendix A: Quick Navigation Reference

I want to...	Go to...
Watch a live transaction	Runtime Operations → Transaction Workflow Screen
Run a new test transaction	Transaction Workflow Screen → green Play button
Search past transactions	Transaction Workflow Screen → Search bar
Approve a paused transaction	Transaction Workflow Screen → current step card → Approve
See failed reversals	Transaction Workflow Screen → Reversal Recovery Queue
Draw a new workflow	Designer Studio → Workflow Designer
Fix unwired workflow steps	Workflow Designer → link icon (Wiring Audit)
Author a business rule	Designer Studio → Business Rules
Build a formula	Designer Studio → Calculation Engine
Configure an external API	Designer Studio → API Designer
Browse ISO 20022 fields	Master Data → ISO Field Registry
Check system health	Runtime Operations → Global Technical Dashboard
Set package + product context	Click package name in top-left corner

Appendix B: Build History

Date	Developer	What was built
Jun 12	Gemini	Infrastructure: FastAPI, SQLite, all data models, seed pipeline
Jun 13–15	Gemini	Full backend for all studios. Screen Designer backend complete.
Jun 16	Gemini	Glassmorphism UI. Package dashboard. Onboarding flow.
Jun 17–18	Gemini	Workflow Designer visual canvas with ReactFlow. Node properties drawer.
Jun 19	Gemini + Claude	ISO Field Registry upgrade. 360° Dashboard redesign. Two-Key Cockpit. 5 studio rebuilds.
Jun 20	Gemini	Calculation Engine rebuild. Message Queue infrastructure. ISO 20022 Template Library. Integration Gateway.
Jun 21	Gemini + Claude	Swimlanes, image-to-workflow, 21 node types. Integration audit. Engine wiring. UX fixes. Rich seed data.
Jun 22 AM	Claude	Transaction Workflow Screen: metro tracker, action buttons, search, bulk ops, reversal drawer. End-to-end execution proof.
Jun 22 PM	Claude	Wiring Audit panel. 137 steps wired. Reversal Recovery Queue endpoint.
Jun 22 Eve	Gemini	Entitlements enforcement (in progress).